“Can you look into the sales data?” is one of the most common requests an analyst receives, and it is not yet a question. Answering it exactly as written is how a week disappears.

CHAPTER 3 — THE DATA PROCESS

Defining the Problem and Asking Questions

Lesson 1 of 6

BY THE END

You can turn a vague request into a specific analytical question.

PLATE I — THE PROCESS, QUESTION TO DECISION



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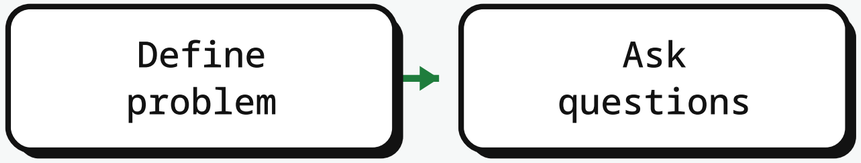


PLATE I — THE PROCESS, QUESTION TO DECISION



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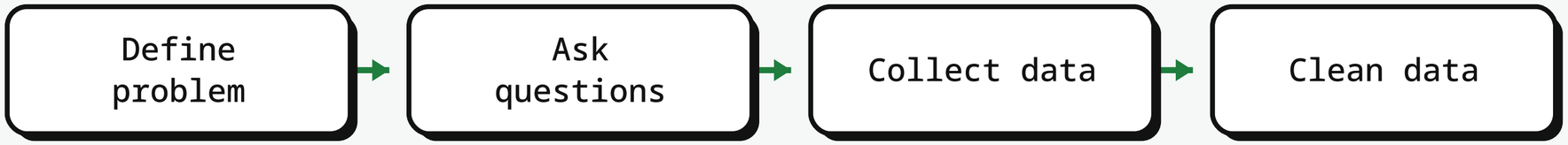


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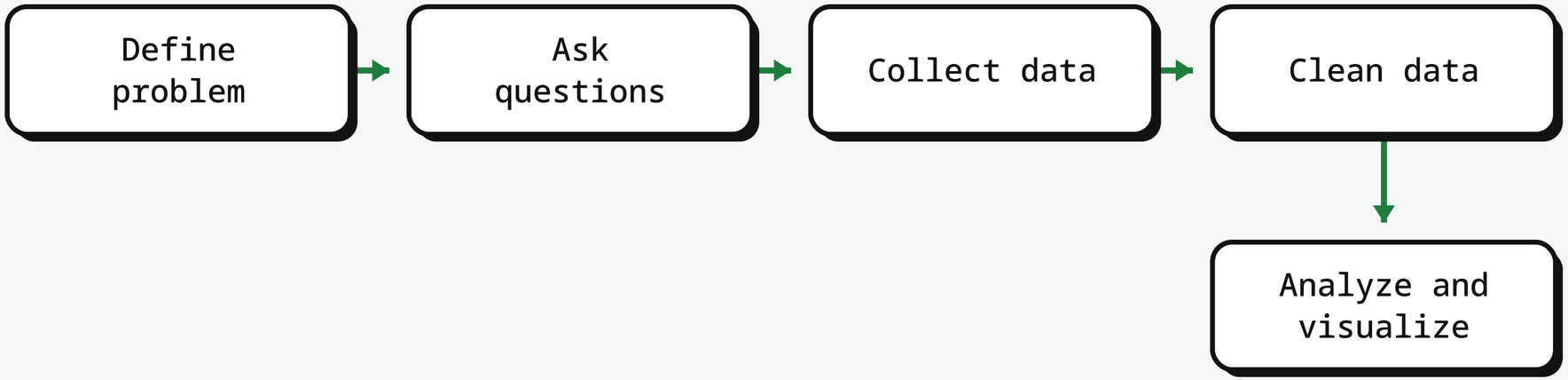


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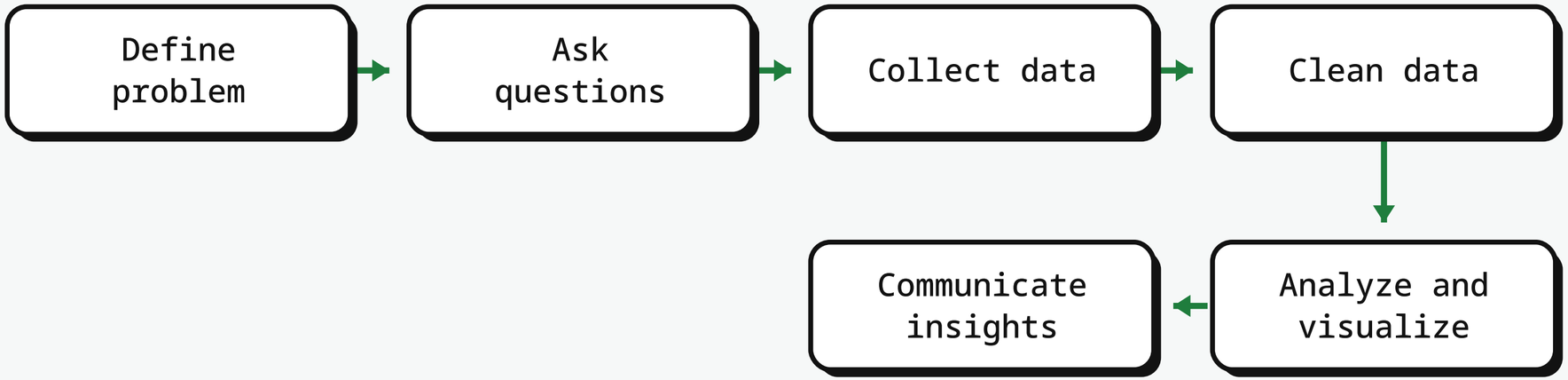
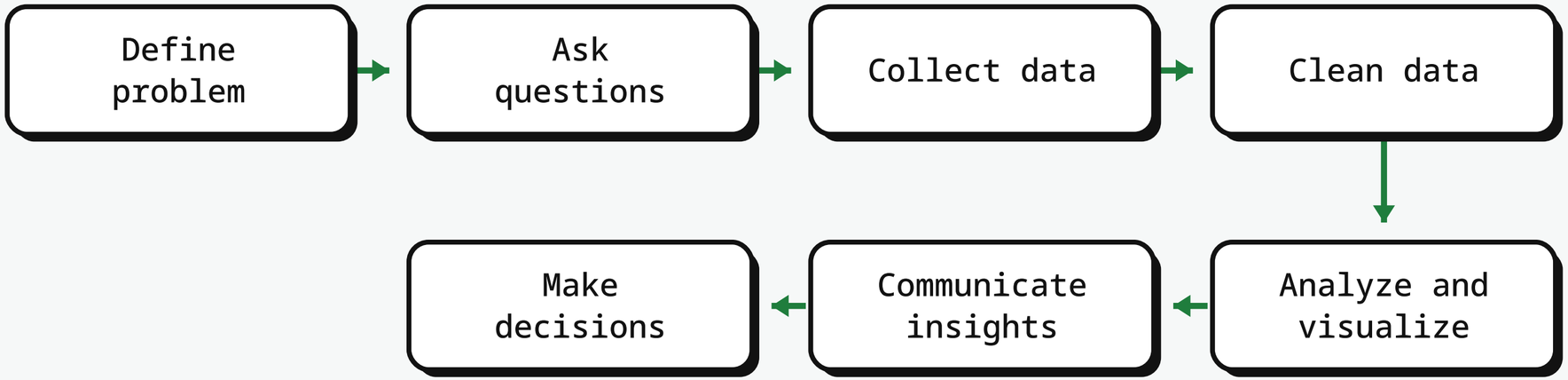


PLATE I — THE PROCESS, QUESTION TO DECISION



every stage depends on the question you asked first

Start with the business problem

A common mistake is jumping straight into data without knowing what you are trying to solve. Always begin by asking:

- What problem are we trying to solve?
- Who cares about the answer?
- What decision will this analysis support?

Start with the business problem

EXAMPLE

- Weak question: “Let’s analyze the sales data.”
- Strong question: “Why did sales drop in Burco last quarter, when Hargeisa and Muqdisho held steady?”

FOR A LOCAL SHOP, THE SAME IDEA APPLIES

- Weak question: “Let’s look at the sales spreadsheet.”
- Strong question: “Why did rice sales drop in the last two weeks compared to the month before?”

Know your stakeholders

A **stakeholder** is anyone who needs the results of your analysis.

EXAMPLES

- A manager deciding on marketing spend
- A product team improving an app feature
- A finance team reviewing budget performance

UNDERSTANDING STAKEHOLDERS HELPS YOU

- Ask the right questions
- Choose relevant metrics
- Present results in a useful way

How to ask good analytical questions

GOOD QUESTIONS ARE

- `Specific` — focused on one issue
- `Measurable` — answerable with data
- `Relevant` — connected to a real decision
- `Clear` — easy for others to understand

EXAMPLES

- `Which product had the highest return rate last month?`
- `Are new customers more likely to stop ordering within their first month?`
- `Which of our three cities is slowest to pay its invoices?`

4 — WORKED EXAMPLE

A training centre in Hargeisa notices that fewer students are enrolling.

STAKEHOLDERS

the centre manager and the person who runs recruitment

PROBLEM

enrolment is falling

QUESTIONS

- When did enrolment start dropping?
- Did it drop across every course, or only some?
- Are fewer people asking about courses, or are the same people asking and not signing up?

KEY TERMS

TERM	DEFINITION
Business problem	The issue or decision the analysis is meant to support
Stakeholder	A person or group affected by, or interested in, the results of the analysis
Analytical question	A clear question that data can help answer